

Project Specification: Ascend Services (v6.0)

1. Project Overview & Architecture

Ascend Services is a ministry-focused web platform designed to foster community connection by bridging the gap between individuals in need and volunteers eager to serve. The version 6.0 release focuses on robust security, automated service level monitoring, and high portability.

Technical Foundations

- **Hosting & Portability:** The application is containerized via Docker to ensure cloud-agnostic deployment, with AWS ECS/EC2 as the primary production environment.
- **Authentication Engine:** A "passwordless-first" strategy using Passkeys (WebAuthn). To ensure inclusivity for users on legacy hardware, Email Magic Links serve as the mandatory secondary fallback.
- **Observability:** Integrated error tracking and structured JSON logging are required. A dedicated `/api/health` endpoint must verify database connectivity to facilitate AWS ALB/ECS health checks.
- **Preferred Tech Stack:**
 - **Backend:** C# / .NET Core or Next.js (Full-stack).
 - **Database:** PostgreSQL with Entity Framework or Prisma.
 - **Styling:** Tailwind CSS.

2. Branding & Design System

The visual identity of Ascend Services should communicate warmth, reliability, and compassion.

- **Tone:** Uplifting, welcoming, trustworthy, and compassionate.
- **Color Palette:** Subdued earth tones, soft sky blues, and warm sun yellows.

- **Logo Design:** The logo consists of three lines swooping upwards, paired with a larger arrow pointing primarily (though not vertically) upwards.
- **Navigation:** A persistent top menu featuring: Home, Volunteer to Serve, Request Help, Help Wanted Board, Admin Dashboard, and Profile.

3. Database Schema

Users Table (Servers & Admins)

Field	Type	Notes
id	UUID	Primary Key
name	String	Full Name
email	String	Unique; login identifier
role	Enum	SERVER or ADMIN
services_provided	Array	List of skills/services
onboarding_token	String	Magic link token for CSV imports

Requests Table (Service Requests)

Field	Type	Notes
id	UUID	Primary Key
requester_name	String	Contact name
service_type	String	Category of help needed
neighborhood	String	Public-facing location
street_address	String	Private; visible only when assigned

status		NEW, ASSIGNED, CONTACT_MADE, IN_PROGRESS, COMPLETE
tracking_token	String	Unique; for unauthenticated self-service

Audit & Settings

- **AuditLog Table:** Captures timestamps, actor IDs, request IDs, and state transitions (e.g., Status Changes) for transparency and accountability.
- **SystemSettings Table:** Configurable thresholds for SLA monitoring, specifically `unassigned_alert_hours`, `stalled_contact_alert_hours`, and `stalled_progress_days`.

4. Automated Workflows & SLA Monitoring

Bulk Import & Upsert Pipeline

The Admin interface supports `.csv` or `.xlsx` uploads.

- **Upsert Logic:** If an email exists, the system appends new skills to the user's profile and notifies them.
- **New Users:** The system generates a secure `onboarding_token` and triggers a "Welcome" email containing a registration magic link.

Connection Email

Upon a request being moved to `ASSIGNED`, the system automatically emails the Requester. This email introduces the volunteer by name, includes their bio, and BCCs Admins for oversight.

SLA Monitoring (Cron Job)

The system actively monitors for delays in service:

- **Unassigned Alert:** Triggers if a **NEW** request remains unassigned beyond the configured threshold.
- **Stalled Assignment:** Triggers if an **ASSIGNED** request does not progress to **CONTACT_MADE** within the specified timeframe.
- **Stalled Progress:** Triggers if a request remains **IN_PROGRESS** longer than the allowed duration.
- **Result:** Admin notification and an **SLA_BREACH** entry in the observability logs.

5. Page-by-Page Specifications

Public Facing Pages

- **Welcome/Home:** Features an uplifting hero section and the ministry mission statement. Clear CTAs: "I Want to Serve" and "I Need Help."
- **Register to Serve:** Captures volunteer details and triggers immediate Passkey registration or Magic Link delivery.
- **Request Help:** Simple form for those in need. Includes location details and a prayer request area with privacy toggles.
- **Requester Self-Service Portal:** A private, token-based view for requesters to track status or cancel requests without an account.

Authenticated Hubs

- **Volunteer Hub (Help Wanted Board):**
 - **Tab 1:** A "Sticky Note" style board showing **NEW** requests, filtered by the user's specific skills.
 - **Tab 2:** "My Assignments" for active tasks, including a fallback "I Can No Longer Help" button.
- **Admin Dashboard:** Comprehensive management of requests and users. Includes the Bulk Import zone and global SLA settings.
- **First-Time Onboarding:** A mandatory flow for imported users to set up their Passkey and complete their professional bio.

6. Engineering Implementation Roadmap

1. **Core Scaffolding:** Initialize PostgreSQL data models and AuditLog interfaces.

2. **Auth Implementation:** Configure WebAuthn/Passkey providers and fallback Magic Link logic.
3. **UI Development:** Build the Admin Grid using TanStack Table or MudBlazor and implement the "Sticky Note" volunteer board.
4. **Backend Services:** Develop the CSV processing pipeline with staging/preview logic and the Cron-based SLA monitor.
5. **Deployment:** Deliver a production-ready `Dockerfile` and `docker-compose.yml` with structured logging enabled.

Approved By: Person

Effective Date: Date